Rest-Tuning: Data-Efficient Adaptation of EEG Foundation Models to Individuals via Resting-State Signals

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Abstract

EEG foundation models (FMs) offer strong population-level priors but degrade on individual subjects due to anatomy-driven distribution shifts. We propose **Rest-Tuning**, a task-agnostic calibration framework that personalizes a population FM using just 1–3 minutes of unlabeled resting-state EEG. Rest-Tuning employs masked teacher-student alignment to produce a subject-calibrated backbone, reused as initialization across multiple tasks from the same subject. Across two multi-subject datasets and four EEG FMs, Rest-Tuning outperforms task-only adaptation, with up to 13.3% absolute gain in downstream task accuracy and over 40% average reduction in labeled task data. Calibration saturates quickly and requires under 3 minutes of compute per subject. Representation analyses show reduced subject-to-population distance and smoother LoRA adapter interpolation paths, indicating improved compatibility between rest- and task-adapted solutions. These results establish resting-state EEG as a **practical, fast, and reusable calibration interface**, enabling population EEG FMs to function as reliable, subject-specific BCIs and serve as reusable initialization for downstream task adaptation.

1 Introduction

Electroencephalography (EEG) is widely used for analyzing and decoding diverse brain states due to its non-invasive, portable, and temporally precise nature. In practice, however, inter-subject variability often exceeds task-related differences [1, 2], causing models trained on one group of subjects to misinterpret subject-specific patterns as task-relevant signals. This distribution mismatch limits model transferability and generalization to unseen individuals (Figure 1a), making the mitigation of cross-subject discrepancies a central challenge in EEG research [3–6].

Pretraining has been shown to improve out-of-distribution generalization by learning more robust and transferable representations [7, 8]. In EEG, recent foundation models similarly enhance cross-subject transfer and downstream decoding performance [9–15]. However, EEG signals exhibit stronger and more complex distribution shifts than other modalities: inter-subject variability arises from anatomy, physiology, electrode coupling, impedance, and cognitive state [3–6], which often causes pretrained EEG foundation models to fail on individual subjects. Achieving reliable performance typically requires large amounts of labeled data and model optimization for each subject–task pair (Figure 1b), a limitation that is especially pronounced in real-world BCI scenarios where systems are rarely restricted to a single fixed task.

Resting-state EEG, which contains stable subject-specific features across spectral, spatial, and connectivity patterns [16–20], provides a natural substrate for low-cost adaptation. Existing subject calibration methods, while effective, typically rely on rest-task paired data, which are costly to collect

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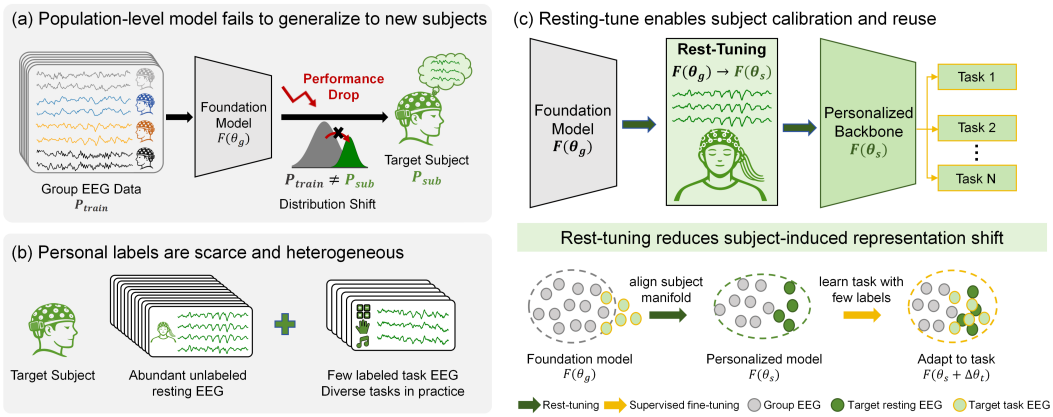


Figure 1: **Overview of Rest-Tuning.** (a) A population EEG FM F_{θ_g} can fail on a new subject because the target-subject distribution differs from the pretraining population. (b) Labeled task EEG is scarce and task-specific, whereas resting-state EEG is comparatively easy to acquire and task-agnostic. (c) Rest-Tuning uses resting EEG to calibrate the backbone into F_{θ_s} before downstream supervision. The same subject-calibrated backbone can then be reused across multiple tasks with lightweight task-specific heads or adapters.

and must be repeated for each new downstream task [21–23]. In addition, task-specific adaptation often requires large amounts of labeled trials, limiting practical scalability and real-world deployment (Figure 1b). These challenges motivate Rest-Tuning, a label- and task-agnostic framework that personalizes a pretrained EEG FM by fine-tuning a student network to match the representations of a frozen teacher on a few minutes of resting-state EEG, using masked reconstruction and contrastive alignment. This produces a reusable, subject-calibrated backbone that can efficiently support multiple downstream tasks (Figure 1c).

Rest-Tuning improves both uni- and multi-task decoding across datasets and EEG FMs. A single resting calibration also improves data efficiency by reducing the amount of labeled task EEG needed across downstream tasks. Further representation and adapter analyses show that Rest-Tuning reshapes subject-specific feature geometry and improves compatibility between rest- and task-adapted solutions. Our contributions are:

- **Task-agnostic subject calibration.** Introduce Rest-Tuning, a subject-specific adaptation framework producing a reusable backbone for multiple downstream tasks, improving FM performance at the individual level.
- **Individualized rest–task alignment.** Rest-Tuning reduces individual distribution shift, improving within-subject task separability and facilitating multi-task adaptation.
- **Data- and time-efficient adaptation.** Achieve effective personalization with only 1–3 minutes of resting EEG, substantially reducing dependence on labeled task trials.

2 Related Work

EEG foundation models and lightweight decoders. Compact EEG decoders such as EEGNet [24] and convolutional/transformer hybrids [25, 26] remain attractive because they are efficient and often strong under single-task calibration. EEG FMs instead aim to learn reusable population priors from heterogeneous recordings, using architectures such as spatiotemporal tokenization [13] and structured spatial–temporal modeling [12, 14, 15]. These models provide stronger population priors, but the best adaptation strategy for new individuals remains unresolved.

Subject personalization and EEG domain adaptation. EEG adaptation methods address shifts across subjects, sessions, devices, and tasks. Recent FM-oriented work studies pretraining–downstream mismatch and physiological-signal personalization, for example through test-time self-supervision or efficient adapters [27, 6, 5]. More broadly, EEG personalization methods often treat each subject as a target domain and adapt directly on the downstream task, using labeled

target-subject trials, meta-learning, or task-specific assumptions [28, 29]. Baseline or resting EEG has also been used to suppress subject-specific nuisance variation or assist motor-imagery transfer, but these methods remain tied to a particular task or model [21–23]. Rest-Tuning differs by using unlabeled resting EEG as a task-agnostic subject calibration stage for pretrained EEG FMs before any downstream task is specified.

Resting-state EEG and task-state relationships. Resting-state organization captures trait-like neural structure that constrains task responses. Prior work has related intrinsic connectivity to task activation, information transfer, and behavioral performance, suggesting that rest signals carry subject-specific structure relevant to later task states [16–19]. These findings motivate resting EEG as a calibration signal, but they do not directly establish how to use rest data to personalize modern EEG FMs. Our work tests this link computationally by asking whether aligning an FM to a subject's resting distribution improves downstream task adaptation across multiple tasks and backbones.

3 Method

3.1 Problem setup

Our goal is to obtain a reusable subject-calibrated backbone for EEG FMs that can support multiple downstream tasks with limited labeled data.

Let F_{θ_g} be a pretrained EEG FM with population parameters θ_g. For a target subject s, we assume access to unlabeled resting-state EEG windows

$$\mathcal{D}^s_{\text{rest}} = \{x^r_i\}_{i=1}^{N_r}, \quad x^r_i \in \mathbb{R}^{C\times T}, \tag{1}$$

and limited labeled task data for one or more downstream tasks $\mathcal{T}$,

$$\mathcal{D}^{s,\mathcal{T}}_{\text{task}} = \{(x^t_j, y^t_j)\}_{j=1}^{N_t}, \quad N_t \ll N_{\text{population}}. \tag{2}$$

Rest-Tuning follows a two-stage personalization pipeline. First, an unlabeled subject-calibration stage maps the population parameters θ_g to subject-calibrated parameters θ_s using only $\mathcal{D}^s_{\text{rest}}$. Second, downstream task adaptation starts from F_{θ_s} and uses the labeled task data $\mathcal{D}^{s,\mathcal{T}}_{\text{task}}$. This separation is central to our evaluation: the task-only baseline uses the same downstream labels and adaptation interface, but starts from the original population model F_{θ_g} instead of the Rest-Tuned model F_{θ_s}.

3.2 Rest-Tuning

Teacher-student architecture. Rest-Tuning adapts the FM using a masked teacher-student objective on resting windows. Given a resting segment x^r_i, we construct a masked view

$$\tilde{x}^r_i = \mathcal{M}_\rho(x^r_i), \tag{3}$$

where $\mathcal{M}_\rho$ masks channels or temporal patches with ratio ρ. A trainable student, initialized from θ_g, encodes the masked view, while a frozen teacher keeps the original population parameters and encodes the unmasked view:

$$z^{(st)}_i = g\left(F_\theta(\tilde{x}^r_i)\right), \qquad z^{(te)}_i = \text{sg}\left[g\left(F_{\theta_g}(x^r_i)\right)\right], \tag{4}$$

where $g(\cdot)$ is a projection head and sg denotes stop-gradient. The frozen teacher anchors the calibration to the pretrained population prior, which is important because each subject provides only a small amount of rest data. Further details are provided in Appendix G.

Loss function. For a mini-batch of n resting windows ($n = 32$), we compute normalized embeddings $\hat{z}^{(st)}_i$ and $\hat{z}^{(te)}_j$ and form the student-teacher similarity matrix

$$S_{ij} = \hat{z}^{(st)\top}_i \hat{z}^{(te)}_j. \tag{5}$$

Each masked view is matched only to its own unmasked teacher view, giving an identity target distribution over the mini-batch. The contrastive loss is

$$\mathcal{L}_{\text{rest}} = -\frac{1}{n}\sum_{i=1}^{n} \log \frac{\exp(S_{ii}/\tau)}{\sum_{j=1}^{n}\exp(S_{ij}/\tau)}. \tag{6}$$

Here, τ is the contrastive temperature. The masked contrastive form follows standard self-supervised representation learning, but its role here is different: it is applied as a per-subject, task-agnostic calibration stage rather than as population pretraining or downstream task adaptation [30–32]. To stabilize calibration and reduce overfitting to short rest recordings, we add a teacher–student consistency regularizer

$$\mathcal{L}_{\text{mse}} = \frac{1}{n} \sum_{i=1}^{n} \left\| \hat{z}_i^{(st)} - \hat{z}_i^{(te)} \right\|_2^2, \tag{7}$$

with total loss

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rest}} + \lambda_{\text{mse}} \mathcal{L}_{\text{mse}}. \tag{8}$$

We set $\lambda_{\text{mse}} = 0.1$ in all experiments. Optimizing $\mathcal{L}_{\text{total}}$ on $\mathcal{D}_{\text{rest}}^{s}$ produces the subject-calibrated parameters θ_s. Detailed hyperparameter settings and ablation results are provided in Appendix C.The implementation of Rest-Tuning is available at an anonymous repository: `https://anonymous.4open.science/r/Rest-Tune-71DE/`.

3.3 Subject calibration and task adaptation interfaces

We perform subject-specific calibration using Rest-Tuning with two parameterizations that balance capacity and efficiency: (i) Rest-Full, which updates all backbone parameters and stores a full subject-specific checkpoint; (ii) Rest-LoRA, which inserts low-rank adapters into attention blocks while freezing the pretrained backbone, storing a compact subject-specific adapter. These Rest-Tuned models serve as the initialization F_{θ_s} for downstream task adaptation.

For downstream tasks, we evaluate three adaptation interfaces: (1) Linear probing, which freezes the calibrated backbone and trains a task head; (2) Task-LoRA, which trains task-specific low-rank adapters; (3) Task-Full, which performs full supervised fine-tuning. All task-only baselines employ the same task labels and interfaces but omit the Rest-Tuning stage. When Task-LoRA or Task-Full further modifies parameters for a given task, the original Rest-Tuned checkpoint is retained and reused as the initialization for future tasks from the same subject.

3.4 Subject-to-group alignment analysis

To test whether Rest-Tuning reduces the mismatch between an individual subject and the population representation, we perform a leave-one-subject-out alignment analysis. Since the original pretraining distribution is unavailable, we approximate the population prior within each dataset using the remaining subjects. For each held-out subject s_i, we first adapt the pretrained backbone on task EEG from the other $N-1$ subjects using the same self-supervised pretraining objective, excluding all data from s_i. This yields a subject-excluded population model $F_{\theta_{-i}}$.

In the baseline condition, both the held-out subject and the remaining subjects are encoded by $F_{\theta_{-i}}$:

$$P_{-i}^{\text{base}} = F_{\theta_{-i}}(X_{-i}), \qquad P_{i}^{\text{base}} = F_{\theta_{-i}}(X_i),$$

where X_i denotes EEG segments from the held-out subject and X_{-i} denotes EEG segments from the remaining subjects.

We then apply Rest-Tuning to $F_{\theta_{-i}}$ using only the held-out subject's unlabeled resting EEG, obtaining a subject-calibrated encoder $F_{\theta_i^{\text{rest}}}$. Both the held-out and population embeddings are re-encoded with this same encoder:

$$P_{-i}^{\text{rest}} = F_{\theta_i^{\text{rest}}}(X_{-i}), \qquad P_{i}^{\text{rest}} = F_{\theta_i^{\text{rest}}}(X_i).$$

The subject-to-population mismatch is quantified by the Wasserstein distance:

$$D_i^{\text{base}} = W(P_i^{\text{base}}, P_{-i}^{\text{base}}), \qquad D_i^{\text{rest}} = W(P_i^{\text{rest}}, P_{-i}^{\text{rest}}), \tag{9}$$

where $W(\cdot,\cdot)$ denotes the Wasserstein distance between embedding distributions. We repeat this procedure for every held-out subject. A decrease from D_i^{base} to D_i^{rest} indicates that Rest-Tuning moves the held-out subject closer to the population embedding geometry using only resting-state EEG.

Table 1: **Uni- and multi-task** performance under 70% training data.

EEG Foundation Model	Rest-Tuning	Task-SFT	Trainable Params (Rest/Task)	M3CV				COGBCI			
				Unitask		Multitask		Unitask		Multitask	
				$\overline{\text{ACC}}$(%)	ΔACC(%)	$\overline{\text{ACC}}$(%)	ΔACC(%)	$\overline{\text{ACC}}$(%)	ΔACC(%)	$\overline{\text{ACC}}$(%)	ΔACC(%)
LaBraM	/	Task-LoRA	0 / 5.00%	56.96	/	36.04	/	60.46	/	56.73	/
	Rest-LoRA	Task-LoRA	5% / 5%	64.34	↑7.38	41.94	↑5.90	63.53	↑3.07	61.20	↑4.47
	Rest-Full	Task-LoRA	100% / 5%	64.74	↑7.78	42.90	↑6.86	68.94	↑8.48	66.71	↑9.98
	/	Task-Full	0 / 100%	61.66	/	45.95	/	72.58	/	73.44	/
	Rest-Full	Task-Full	100% / 100%	68.33	↑6.67	49.79	↑3.84	75.52	↑2.94	78.13	↑4.69
BIOT	/	Task-LoRA	0 / 4.5%	65.56	/	50.04	/	90.14	/	88.94	/
	Rest-LoRA	Task-LoRA	4.5% / 4.5%	70.57	↑5.01	52.24	↑2.20	90.98	↑0.84	89.50	↑0.56
	Rest-Full	Task-LoRA	100% / 4.5%	68.02	↑2.46	50.59	↑0.55	91.21	↑1.07	89.58	↑0.64
	/	Task-Full	0 / 100%	67.46	/	50.74	/	90.39	/	89.38	/
	Rest-Full	Task-Full	100% / 100%	69.27	↑1.81	52.82	↑2.08	91.24	↑0.85	90.45	↑1.07
CBraMod	/	Task-LoRA	0 / 3.03%	48.02	/	24.18	/	47.76	/	38.63	/
	Rest-LoRA	Task-LoRA	3.03% / 3.03%	50.46	↑2.44	24.95	↑0.77	60.95	↑13.19	40.24	↑1.61
	Rest-Full	Task-LoRA	100% / 3.03%	55.45	↑7.43	27.35	↑3.17	61.01	↑13.25	50.15	↑11.52
	/	Task-Full	0 / 100%	58.89	/	34.36	/	71.80	/	70.59	/
	Rest-Full	Task-Full	100% / 100%	62.49	↑3.60	37.55	↑3.19	75.40	↑3.60	73.69	↑3.10
EEGPT	/	Task-LoRA	0 / 3.03%	52.03	/	27.62	/	54.77	/	48.59	/
	Rest-LoRA	Task-LoRA	3.98% / 3.98%	55.00	↑2.97	27.96	↑0.34	55.47	↑0.70	49.17	↑0.58
	Rest-Full	Task-LoRA	100% / 3.98%	56.30	↑4.27	28.55	↑0.93	54.99	↑0.22	50.19	↑1.60
	/	Task-Full	0 / 100%	54.14	/	26.81	/	55.05	/	48.14	/
	Rest-Full	Task-Full	100% / 100%	66.10	↑11.96	30.30	↑3.49	57.47	↑2.42	54.69	↑6.55

3.5 Adapter-space interpretation

To analyze whether Rest-Tuning changes the parameter landscape for later task adaptation, we study linear interpolation in LoRA adapter space [33–35]. Let W_0 be the frozen backbone and ΔW_{rest} a Rest-LoRA update. For a downstream task, let ΔW_{task} be the task-supervised LoRA update. We evaluate the rest-to-task path

$$W(\lambda) = W_0 + (1-\lambda)\Delta W_{\text{task}} + \lambda \Delta W_{\text{rest}}, \quad \lambda \in [0,1]. \tag{10}$$

A smoother path and smaller mid-point accuracy drop suggest that the rest adapter lies in a region compatible with task solutions. We also compare task-to-task paths between ΔW_{task_1} and ΔW_{task_2}:

$$W(\lambda) = W_0 + (1-\lambda)\Delta W_{\text{task}_1} + \lambda \Delta W_{\text{task}_2}. \tag{11}$$

These analyses test whether subject calibration makes rest- and task-derived adapters, or adapters from different tasks, more mutually compatible within the same subject.

4 Experiments

We evaluate Rest-Tuning along three axes: performance, representation alignment, and data efficiency. We first assess uni-task and multi-task decoding across datasets and EEG FMs. We then analyze subject-to-group embedding distance, within-subject task separability, and LoRA interpolation to characterize representation alignment. Finally, we test whether one resting calibration reduces label requirements and supports multiple downstream tasks under matched supervision.

4.1 Experimental setup

Datasets. We evaluate on two multi-subject EEG datasets: M3CV [36] and COGBCI [37]. For M3CV, we use 95 subjects with individual labels, covering resting-state, sensory, P300, and motor paradigms. For COGBCI, we use session 1 from 29 participants and focus on NBack and MATB workload tasks. Detailed dataset statistics are reported in Table A1.

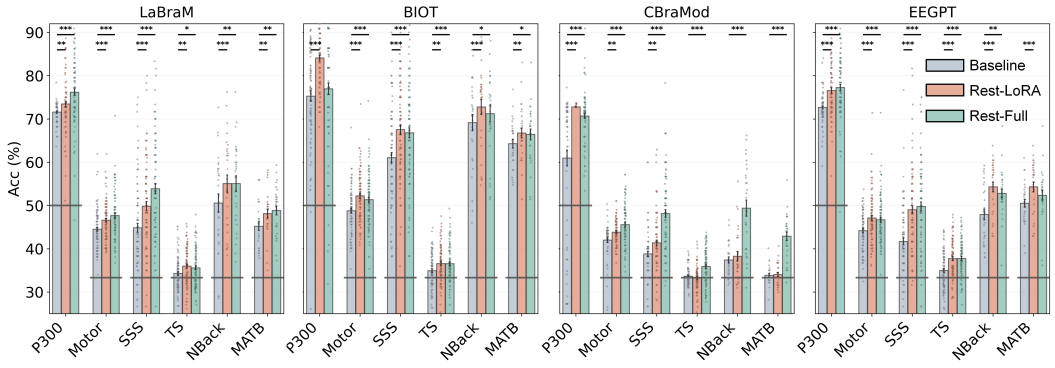


Figure 2: **Linear probing performance.** Bar plots show the uni-task decoding accuracy across six downstream tasks for four EEG FMs. Gray: no-rest baseline; orange: Rest-LoRA; green: Rest-Full. Error bars denote standard deviation. Significance was tested with paired-subject comparisons (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$).

Backbones and adaptation. We evaluate four EEG FMs: LaBraM [12], BIOT [13], CBraMod [15], and EEGPT [14]. Rest-Full updates all backbone parameters with mask ratio $\rho = 0.3$ to preserve sufficient information for stable full-network adaptation, while Rest-LoRA injects rank-8 low-rank adapters into attention blocks and uses a higher mask ratio $\rho = 0.7$ to encourage stronger regularization of the compact adapters. For downstream evaluation, we compare task-only adaptation against Rest-Tuning followed by the same task interface. Unless otherwise stated, results are averaged across target subjects and reported as mean $\pm$ standard deviation.

Protocols. We evaluate uni-task, multi-task decoding, rest-duration sensitivity, accuracy–cost trade-off with lightweight backbones, t-SNE analysis, and LoRA interpolation. Multi-task decoding is defined as merging multiple uni-tasks into a single multi-class problem, designed to simulate the complexity of real-world BCI scenarios. Training ratios include 10% and 70% for the main FM comparison, and 5%, 10%, 50%, and 70% for the lightweight-backbone comparison.

4.2 Rest-Tuning improves decoding across tasks and FMs

We first evaluate whether resting calibration improves downstream decoding while keeping the task interface fixed. Figure 2 reports linear probing accuracy across six downstream tasks and four EEG FMs. Rest-Tuning consistently outperforms the no-rest baseline, indicating that it enhances the backbone representations and improves task separability even before parameter-heavy supervised adaptation.

Table 1 summarizes uni-task and multi-task results for each backbone. Both Rest-LoRA and Rest-Full consistently improve task-only adaptation performance across tasks and backbones, with gains reaching approximately 10% in multiple tasks. These improvements persist across different task interfaces, highlighting that Rest-Tuning serves as a general subject-calibration stage rather than a task-specific adjustment.

All results above are obtained using 70% of the task labels for training. We further evaluate extreme low-label scenarios (train 10%), showing that backbone optimization (linear probing) and task adaptation gains remain robust (see Table A7).

4.3 Rest-Tuning aligns subject representations

Rest-Tuning aligns subject-specific representations with the group prior. Pretrained EEG FMs capture a group-level representation prior, but individual subjects exhibit idiosyncratic neural and signal characteristics that shift their embeddings away from this prior. Rest-Tuning aims to reduce this subject-to-group mismatch before any task-specific labels are used (Figure 3a).

We validate this hypothesis on the COGBCI dataset with LaBraM. Group-level embeddings P_{group} and individual subject embeddings P_{sub} are constructed following the method described in Sec. 3.4. Figure 3b shows pairwise Wasserstein distances across 10 subjects: the lower triangle corresponds

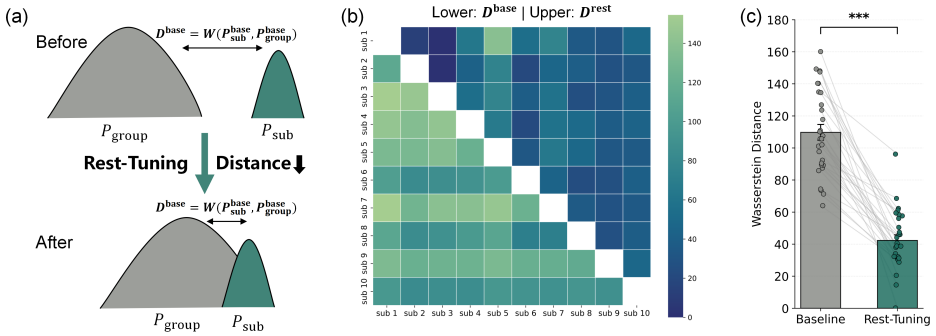


Figure 3: **Rest-Tuning reduces subject-to-group mismatch.** (a) Conceptual illustration: single-subject embedding (P_{sub}, green) moves closer to the group prior (P_{group}, gray) after Rest-Tuning. (b) Pairwise Wasserstein distances for 10 subjects; lower triangle: baseline, upper triangle: Rest-Tuned. (c) Subject-to-group distances before (D_i^{base}) and after (D_i^{rest}) Rest-Tuning. Each dot represents a subject. Paired t-test: *** $p < 0.001$.

to baseline embeddings, and the upper triangle corresponds to Rest-Tuned embeddings. Figure 3c visualizes the subject-to-group distances before (D_i^{base}) and after (D_i^{rest}) Rest-Tuning. Paired t-tests confirm that Rest-Tuning significantly reduces these distances (*** $p < 0.001$), supporting that it effectively aligns individual embeddings with the group-compatible representation space and corrects subject-specific distributional mismatch.

Rest-Tuning improves adapter-space compatibility. Building on the reduction of subject-to-group mismatch, we examine how Rest-Tuning affects task-specific adaptation. By aligning embeddings across subjects, Rest-Tuning creates a smoother, more consistent parameter space, facilitating downstream adaptation. We probe this effect using LoRA adapters to explore the low-loss landscape. Rest-to-task interpolation (Figure 4a–c) shows Rest-Tuning adapters maintain higher accuracy than randomly initialized LoRA, placing them closer to task-relevant low-loss regions. Task-to-task interpolation (Figure 4d–f) reduces performance drops between adapters, demonstrating a smoother, more reliable adapter landscape that facilitates cross-task transfer. Additional trial-level t-SNE visualizations (Figure A5) show that Rest-Tuning enhances task-cluster separability across adaptation stages, indicating more stable and task-informative embeddings.

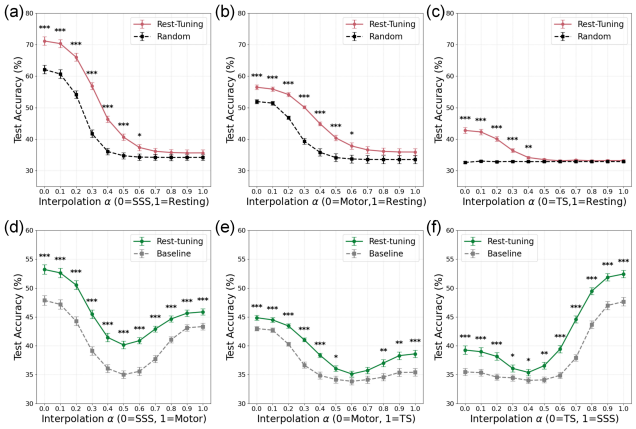


Figure 4: **LoRA adapter interpolation.** (a–c) Rest-to-task interpolation: Rest-Tuned adapters maintain higher accuracy than randomly initialized LoRA. (d–f) Task-to-task interpolation.

4.4 Rest-Tuning reduces supervision and calibration cost

One rest calibration reduces task-label dependence across tasks. We further examine whether Rest-Tuning reduces the need for labeled task trials. Since Rest-Tuning calibrates the backbone with unlabeled resting EEG, the subject-specific representation does not need to be relearned from scarce task labels for each downstream task.

Figure 5 shows that Rest-Tuned models reach baseline accuracy with substantially fewer labeled trials. In the multi-task setting of M3CV, Rest-LoRA and Rest-Full save up to 34.2% and 43.4% of labeled training data, respectively. Across individual tasks and datasets (M3CV and COGBCI), the reduction ranges from 20% to 60%, indicating that a single resting calibration can be reused across multiple task

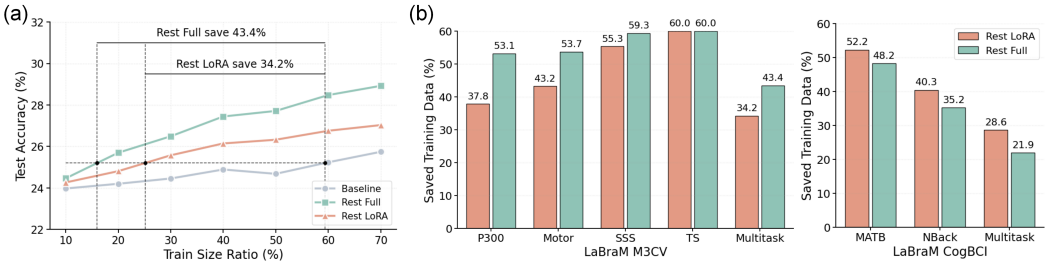


Figure 5: **Rest-Tuning reduces labeled task data requirements.** (a) Test accuracy of LabraM on M3CV multitask under varying training sample ratios for baseline, Rest-LoRA, and Rest-Full. (b) Percentage of saved training data across uni-task and multi-task evaluations in two datasets.

domains. Rest-Full generally provides larger savings, while Rest-LoRA offers a parameter-efficient alternative. These results suggest that Rest-Tuning does not replace task supervision, but substantially reduces task-label dependence by amortizing subject calibration through resting EEG. Since we only tested training fractions from 10% to 70%, the observed maximum savings is 60%. Additional model- and task-specific results are reported in Appendix H.

Rest-Tuning requires only minutes of rest EEG. To assess the amount of resting-state EEG required for practical subject-specific adaptation, we varied the duration of EEG used during Rest-Tuning while keeping all other settings fixed. Figure 6 shows the downstream multitask classification accuracy of LabraM with Rest-Full as a function of resting EEG duration. Accuracy improves rapidly with increasing rest data, reaching near-saturation after approximately 60–120 seconds, with only marginal gains beyond two minutes. These results indicate that 1–3 minutes of resting-state EEG are sufficient to effectively calibrate the subject-specific backbone for multi-task decoding, minimizing the data collection burden while maintaining strong downstream performance.

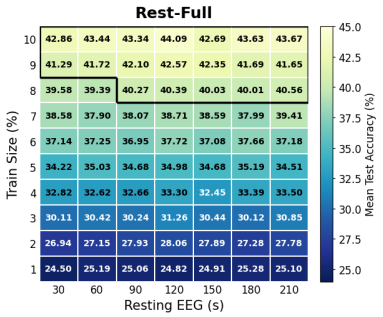


Figure 6: **Resting EEG duration vs. multitask accuracy.** LabraM Rest-Full performance rapidly saturates with 1–3 minutes of resting EEG.

Computational Efficiency of Rest-Tuning We evaluate the computational overhead of Rest-Tuning on the M3CV dataset using 70% of training data. Across LaBraM and BIOT models, adapting a subject-specific backbone via Rest-Tuning requires under 3 minutes on average per subject (see Table A8 for detailed timings). Critically, Rest-Tuning is task-agnostic: a single calibration can be reused across multiple downstream tasks. As shown in Figure 7, the effective per-task cost decreases as the number of tasks increases. When adapting four tasks, the cost relative to Task-SFT drops to 38.3% for LaBraM and 51.8% for BIOT. These results indicate that Rest-Tuning introduces only a modest one-time computational overhead while providing increasing efficiency in multiple downstream tasks BCI scenarios, highlighting its practical applicability.

Accuracy–cost trade-off across lightweight and foundation backbones. Finally, we compare compact EEG decoders and representative Rest-Tuned FMs under the same multi-task splits. Table 2 reports accuracy at four train-size ratios and the adaptation cost at the 70% label setting. EEGNet and EEGConformer are task-only lightweight decoders and do not use Rest-Tuning; they provide a cost baseline. LaBraM and BIOT use Rest-Full followed by Task-Full adaptation. While the Rest-Tuned FMs have comparable or higher adaptation cost than the lightweight decoders, they also achieve substantially higher accuracy, especially in low-label regimes. The additional Rest-Tuning cost is a one-time calibration that can be reused across tasks, and the improved accuracy may justify the additional cost in many applications. Full per-ratio cost and accuracy results are provided in Table A6.

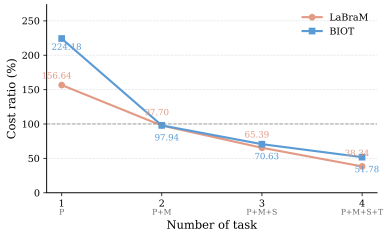


Figure 7: **Effective per-task cost of Rest-Tuning relative to Task-SFT on the M3CV dataset.** The four tasks are P: P300, M: Motor, S: Steady-State Sensory, and T: Transient Sensory.

Table 2: **Backbone accuracy–cost context for multitask decoding.** ACC is reported as mean accuracy (%) across subjects at different labeled train ratios. Cost@70% is the task-adaptation time in seconds at the 70% label setting and includes the Rest-Tuning stage. Lightweight decoders are trained directly on task labels, whereas FM rows use Rest-Full followed by Task-Full adaptation.

Dataset	Model	Protocol	Rest-Tuning Cost (s)	ACC@5%	ACC@10%	ACC@50%	ACC@70%	Cost@70% (s)
COGBCI	EEGNet	Task-only	/	21.47	22.10	62.70	64.38	**287.80**
	EEGConformer	Task-only	/	28.93	34.00	52.77	59.54	1045.37
	LaBraM	Rest+Task-Full	150.37	35.67	44.42	72.85	78.13	528.71
	BIOT	Rest+Task-Full	80.90	**70.41**	**78.56**	**89.32**	**90.45**	369.66
M3CV	EEGNet	Task-only	/	22.08	24.69	34.92	39.38	757.33
	EEGConformer	Task-only	/	24.42	24.14	28.59	32.59	764.47
	LaBraM	Rest+Task-Full	128.67	25.36	28.08	44.08	49.79	**251.23**
	BIOT	Rest+Task-Full	69.72	**30.28**	**36.35**	**47.91**	**52.82**	198.10

5 Discussion and Limitations

We introduced Rest-Tuning, a resting-state calibration interface for adapting EEG FMs to individuals. Our central message is that EEG FMs provide useful population priors, but require a low-cost subject interface for reliable subject-level adaptation. Rest-Tuning provides this interface by using resting EEG to obtain a reusable subject-calibrated backbone before task-specific supervision.

Reusable personalization, not task-free decoding. Rest-Tuning separates subject calibration from task learning. The Rest-Tuned checkpoint is reused as the subject-specific starting point, while task heads are trained with the same labeled data as task-only baselines. Unlike task-specific resting-state adaptation methods such as ResTL, which require separate calibration for each task and multi-subject rest–task pairs, Rest-Tuning uses only target-subject resting EEG to produce a task-agnostic backbone reusable across downstream tasks. Appendix E shows that Rest-Tuning achieves higher accuracy with substantially lower calibration time, supporting its efficiency and effectiveness for subject-adaptive EEG FM personalization.

Representation alignment. The representation analyses indicate that Rest-Tuning reduces subject-specific mismatch in the embedding space. In the subject-to-group distance analysis, Rest-Tuning lowers the Wasserstein distance between target-subject embeddings and the subject-excluded group distribution. In adapter space, LoRA interpolation yields smoother rest-to-task and task-to-task paths, suggesting better compatibility with later task solutions. The t-SNE visualizations provide illustrative evidence of improved within-subject task separability, as shown in Appendix J. These findings support the view that resting EEG provides useful subject information for aligning individual representations with group-level structure.

Calibration cost, label efficiency, and backbone trade-offs. Rest-Tuning adds an unlabeled calibration stage and is therefore not compute-matched to task-only adaptation. Its value is that this one-time subject-calibration cost can be amortized across multiple downstream tasks. This matters most in low-label regimes, where task-only fine-tuning from a population FM is vulnerable to subject shift and overfitting. Our label-efficiency and rest-duration analyses suggest that short resting calibration can reduce downstream label demand, while the lightweight-backbone comparison shows the broader accuracy–cost trade-off: compact EEG decoders remain attractive when computation is the primary constraint, whereas stronger EEG FMs can provide better multitask accuracy under limited labels.

Limitations. Rest-Tuning is a static calibration procedure. EEG is non-stationary, so performance may drift with fatigue, attention changes, or electrode impedance shifts; online or test-time adaptation may be needed for long-term use. Our reusability claim is limited to multiple tasks from the same subject, and broader cross-session or cross-day stability remains to be tested. Rest-Tuning also depends on resting-recording quality: severe artifacts, poor electrode contact, or acquisition mismatch between rest and task recordings may degrade calibration.

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A Datasets and Data Processing

M3CV M3CV is a multi-subject, multi-session, and multi-task EEG benchmark comprising 106 subjects recorded in two sessions, with 15 runs per subject, covering six paradigms and 14 tasks (approximately ∼120,000 epochs in total) [36]. Recordings use a 64-channel EEG setup following the 10–20 system. The paradigms span (i) *resting state* with eyes-open and eyes-closed conditions, (ii) *transient sensory* evoked responses (visual, auditory, and somatosensory), (iii) *steady-state sensory* responses (SSVEP, SSAEP, and SSSEP), (iv) a *visual oddball* paradigm eliciting P300 responses (target vs. non-target), and (v) *motor execution* tasks (e.g., foot, right-hand, and left-hand movements). This wide coverage enables unified evaluation of models across heterogeneous sensory, cognitive, and motor conditions, as well as across recording sessions.

COGBCI COGBCI is an open multi-session, multi-task EEG dataset targeting passive BCI applications, with 29 participants recorded over three sessions spaced one week apart, totaling over 100 hours of EEG [37]. Each session includes (i) *resting state* (eyes-open and eyes-closed) recorded at both the beginning and end of the protocol, and (ii) four cognitive tasks designed to elicit distinct mental states: a *Psychomotor Vigilance Task (PVT)* for vigilance/fatigue, an *N-back* task (0/1/2-back) for graded workload, an arrow-based *Flanker* task for conflict/error processing, and *MATB-II* for more ecological multi-attribute workload manipulation (easy/medium/difficult). EEG is acquired with a 64-channel active-electrode system at 500 Hz (10–20 layout), and the released data are provided in BIDS format with behavioral and subjective measures (e.g., KSS/RSME) for validation and multimodal modeling.

Data Preprocessing To standardize the heterogeneous recordings from M3CV and COGBCI with the input specifications of the EEG FMs, we applied a consistent preprocessing pipeline. Let the raw continuous EEG signal be denoted as $\mathbf{D}_{\text{raw}} \in \mathbb{R}^{C \times L_{\text{raw}}}$, where C represents the channel dimension and f_s the original sampling rate. We first downsampled all signals to a target rate of $f_s' = 200$ Hz and segmented them into fixed-length, non-overlapping windows of duration $T = 1$ second. This yields model inputs $\mathbf{X} \in \mathbb{R}^{C \times N}$, where $N = 200$. To mitigate inter-subject amplitude variability, we applied instance-wise Z-score standardization. For label space construction, we formulated two strategies: for single-task analysis, we preserved the intrinsic local labels y_{local}; for the multi-task setting, to resolve label collisions across different paradigms, we constructed a global label space $\mathcal{Y}_{\text{global}}$ using a strided mapping function Φ. Given a task index i and a stride $\delta > \max_i(K_i)$, the global label is defined as:

$$y_{\text{global}} = \Phi(i, y_{\text{local}}) = i \times \delta + y_{\text{local}}, \tag{12}$$

which projects all task conditions into a unified categorical space, enabling simultaneous evaluation of the personalized backbone across disjoint tasks.

Table A1: Summary of datasets, tasks, classes, and average sample numbers.

Dataset	Subjects	Resting Segments (mean)	Task	Classes	Task Segments (mean)
COGBCI	29	240.0	MATB	3	897.6
			N-Back	3	1042.1
M3CV	95	265.2	Motor	3	814.6
			P300	2	486.5
			Steady-State Sensory	3	332.4
			Transient Sensory	3	198.1

B Foundation Model Summary and Parameter Count

Table A2 summarizes the characteristics of the EEG foundation models used in this work, including model scale, architectural design, pretraining strategy, pretraining signal duration, and supported EEG channels. The four models differ substantially in pretraining scale and channel configurations, ranging from 198 hours to 59,304 hours of pretraining EEG signals and supporting different electrode settings. These differences provide a diverse benchmark for evaluating the effectiveness and generalizability of the proposed Rest-Tuning framework across heterogeneous EEG foundation models.

Table A2: **Summary of recently published brain foundation models.**

Model	Parameters	Architecture / Pretraining Strategy	Pretraining Signal Duration	Supported Channels
BIOT [13]	3.2M	Linear-Transformer / Self-supervised - Contrastive Learning	59,304h	18
EEGPT [14]	25M	CNN-Transformer / Masked Signal Reconstruction	198h	58
LaBraM [12]	5.8M	CNN-Transformer / Masked Signal Reconstruction	2,535h	136
CBraMod [15]	4.0M	CNN-Transformer / Masked Signal Reconstruction	9,246h	Montage-agnostic

Table A3 reports the total and trainable parameters under Full tuning, LoRA tuning, and linear probing. For Full and LoRA tuning, the trainable ratio is calculated with respect to the total parameters of the corresponding tuning strategy. Compared with Full tuning, LoRA reduces the trainable parameters to only 3.03%–5.00%, while linear probing requires only 1,206 trainable parameters for each model.

Table A3: **Total and trainable parameters under different tuning strategies.**

Model	Strategy	Reference Total Params	Trainable Params	Ratio
LaBraM	Full tuning	6,017,936	6,017,936	100.00%
	LoRA tuning	6,334,736	316,800	5.00%
	Linear probing	6,017,936	1,206	0.020%
BIOT	Full tuning	3,186,706	3,186,688	100.00%
	LoRA tuning	3,337,018	150,312	4.50%
	Linear probing	3,186,706	1,206	0.038%
CBraMod	Full tuning	4,924,000	4,924,000	100.00%
	LoRA tuning	5,077,600	153,600	3.03%
	Linear probing	4,924,000	1,206	0.024%
EEGPT	Full tuning	4,924,000	4,924,000	100.00%
	LoRA tuning	5,077,600	153,600	3.03%
	Linear probing	4,924,000	1,206	0.024%

C Model Hyperparameters

Table A4 summarizes the key hyperparameters for Rest-Tuning. Each parameter is chosen to stabilize training, balance efficiency and expressive power, and optimize the calibration of a pretrained EEG foundation model to individual resting-state EEG.

C.1 MSE Weight Ablation Analysis

To evaluate the effect of the MSE constraint, we performed an ablation study on the LaBraM multitask setting using full-task fine-tuning. Table A5 reports the model performance under different λ_{mse} values.

The results indicate:

- Introducing the MSE constraint consistently improves performance compared to removing it ($\lambda_{\text{mse}} = 0$) across training ratios.
- Within the tested range $\lambda_{\text{mse}} \in [0.01, 1]$, model performance is relatively stable and superior to the non-tuned baseline.
- The chosen value $\lambda_{\text{mse}} = 0.1$ achieves the best performance, balancing smoothness of embedding space and gradient signal strength.

These results justify the use of $\lambda_{\text{mse}} = 0.1$ as the default setting for Rest-Tuning in subsequent experiments, ensuring stable adaptation to individual resting EEG while maintaining performance across tasks.

Table A4: Key hyperparameters for Rest-Tuning and their rationale.

Parameter	Value	Update Strategy	Rationale
Mask ratio ρ	0.3(Full)/0.7(LoRA)	Fixed	Full tuning preserves more EEG information; LoRA benefits from stronger masking regularization to enhance feature diversity.
MSE weight λ_{mse}	0.1	Fixed	Ensures embedding space smoothness and amplifies positive-pair gradient.
Temperature τ	$0.1 \rightarrow 0.005$	Cosine annealing	High initial temperature stabilizes early representations; decayed temperature sharpens inter-instance discrimination.
Batch size n	32	Fixed	Ensures sufficient negative samples for contrastive learning; suitable for small resting EEG datasets.
Epoch number	Max 300, early stop patience=30	Early stopping	Prevents overfitting on short resting EEG recordings.
Learning rate	1e-6 $\rightarrow$ 1e-4	Linear warm-up + cosine decay	Warm-up stabilizes early training; cosine decay smooths convergence to final solution.
Optimizer	AdamW	Fixed	Decouples weight decay from gradient update.
LoRA rank	8	Fixed	Low-rank adaptation balances parameter efficiency and expressive capacity.
LoRA alpha	32	Fixed	Scales the residual update to control magnitude of LoRA adaptation.
LoRA dropout	0.1	Fixed	Regularizes LoRA adapters to reduce overfitting.

Table A5: Performance comparison under different λ_{mse} settings (%).

Dataset	Training Ratio	λ_{mse}				
		0	0.01	0.1	1	Baseline
COGBCI	Train 10%	41.31±8.72	43.91±9.83	**44.42±9.82**	42.99±9.15	39.68±7.44
	Train 70%	77.03±10.07	77.88±9.01	**78.13±9.45**	77.28±9.46	73.44±9.53

D Lightweight-backbone accuracy-cost comparison

We include supplementary tables for the lightweight-backbone accuracy-cost comparison.

E Comparison with Other Resting-State Calibration Methods

We further compare Rest-Tuning with ResTL [22], a recent resting-state subject-adaptive transfer method, using the official implementation. Due to the computational cost of ResTL, we evaluate on five randomly selected target subjects under the same downstream task split used for Rest-Tuning. We report mean training time and test accuracy averaged over the five subjects. EEG foundation model(LaBraM and BIOT) with Rest-Tune achieves higher accuracy while requiring only target-subject resting EEG and substantially less training time, as shown in Table A7.

F Rest-Tuning and Task-SFT running time conparison

We report the computational runtime of individual Rest-Tuning and Task-SFT for the LaBraM and BIOT models on the M3CV and COGBCI datasets. The results are summarized in Table A8.

Table A6: Full accuracy and cost comparison for multitask decoding. Cost is reported in seconds as mean ± standard deviation. ACC is reported as mean ± standard deviation across subjects. Bold indicates the best ACC within each train-size setting.

Dataset	Model	5%		10%		50%		70%	
		Cost	ACC	Cost	ACC	Cost	ACC	Cost	ACC
COGBCI	EEGNet	12.06 ± 2.17	0.2147 ± 0.0339	21.87 ± 2.30	0.2210 ± 0.0257	237.14 ± 0.98	0.6270 ± 0.1550	287.80 ± 0.54	0.6438 ± 0.1490
	EEGConformer	53.63 ± 8.48	0.2893 ± 0.0642	115.20 ± 7.76	0.3400 ± 0.1007	685.24 ± 3.67	0.5277 ± 0.1136	1045.37 ± 1.85	–
	LaBraM	63.33 ± 13.01	0.3567 ± 0.0688	90.40 ± 22.76	0.4442 ± 0.0911	323.31 ± 84.08	0.7285 ± 0.0972	378.34 ± 119.40	0.7813 ± 0.0945
	BIOT	85.41 ± 32.46	**0.7041 ± 0.0758**	92.28 ± 37.44	**0.7856 ± 0.0612**	336.35 ± 96.97	**0.8932 ± 0.0373**	288.76 ± 102.02	**0.9045 ± 0.0398**
M3CV	EEGNet	51.50 ± 7.47	0.2208 ± 0.0344	96.98 ± 7.04	0.2469 ± 0.0312	515.89 ± 2.95	0.3492 ± 0.0776	757.33 ± 1.47	0.3938 ± 0.0866
	EEGConformer	29.29 ± 6.65	0.2442 ± 0.0147	55.52 ± 6.65	0.2414 ± 0.0196	410.75 ± 3.81	0.2859 ± 0.0705	764.47 ± 1.97	0.3259 ± 0.0967
	LaBraM	35.14 ± 12.25	0.2536 ± 0.0271	24.15 ± 8.41	0.2808 ± 0.0385	107.42 ± 41.30	0.4408 ± 0.0728	122.56 ± 32.79	0.4979 ± 0.0771
	BIOT	28.72 ± 10.54	**0.3028 ± 0.0402**	39.92 ± 16.90	**0.3635 ± 0.0505**	115.54 ± 42.79	**0.4791 ± 0.0611**	128.38 ± 56.40	**0.5282 ± 0.0630**

Table A7: Comparison of Different Methods on Rest-Task Transfer Learning

Method	Additional Training Data	Training Time	Test Data	$\overline{ACC}$ (%)
ResTL-EEGNet	28-subject rest-task pairs + target rest	128.42 min	Target rest	25.44
ResTL-Conformer	28-subject rest-task pairs + target rest	64.02 min	Target rest	27.33
LaBraM with Rest-Full	target-subject rest only	2.68 min	Target task	83.83
BIOT with Rest-Full	target-subject rest only	4.76 min	Target task	91.64

G Teacher-Student Architecture for Rest-Tuning

Figure A1 illustrates the teacher-student architecture used in Rest-Tuning. Resting-state EEG segments are divided into temporal windows $t_1, t_2, \ldots, t_n$ and optionally masked to create $\tilde{x}_i^r$ for the student network. The **student network** is trainable and receives masked EEG inputs, learning to encode features that are robust to missing channels or temporal patches. Simultaneously, the **teacher network** is frozen and encodes the unmasked EEG, providing a stable reference representing the population prior.

Student embeddings $z_i^{(st)}$ are aligned with teacher embeddings $z_i^{(te)}$ via a contrastive InfoNCE loss, ensuring that each masked student embedding is pulled towards its corresponding teacher embedding while being repelled from other embeddings in the mini-batch. Additionally, a mean-squared error consistency regularizer $\mathcal{L}_{\text{mse}}$ is applied to further smooth the embedding space and stabilize training.

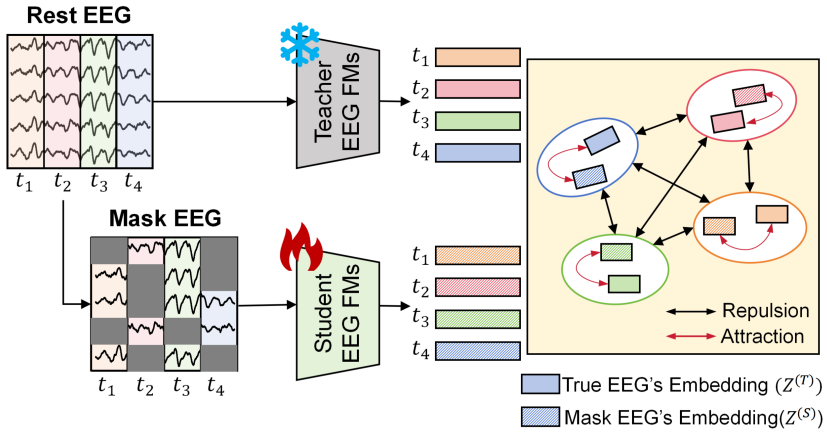


Figure A1: Illustration of the Rest-Tuning teacher-student architecture. Rest EEG segments are masked and fed into the trainable student network, while the frozen teacher network encodes the unmasked EEG. The student embeddings are aligned with the teacher embeddings via a contrastive loss and a consistency regularizer. This figure clarifies how masked EEG views and true EEG embeddings interact during calibration.

Table A8: **Runtime cost comparison of Rest-Tuning and Task-SFT.**

Dataset	Task	LaBraM		BIOT	
		Rest-tune (s)	Task-SFT (s)	Rest-tune (s)	Task-SFT (s)
M3CV (n=95)	P300	128.67	80.60	69.72	31.10
	Motor		51.10		40.08
	SSS		65.07		27.53
	TS		138.87		35.94
	Multitask		122.56		128.38
COGBCI (n=29)	MATB	150.37	208.18	80.90	178.94
	NBack		121.02		164.36
	Multitask		378.34		288.76

H Data-Efficiency Analyses

Figures A2 and A3 provides task-level data-efficiency curves for Baseline, Rest-Full, and Rest-LoRA for both LaBraM and BIOT models. The results show that Rest-Tuning often reaches the same accuracy as the task-only baseline using fewer labeled training samples. This trend is observed across both COGBCI and M3CV tasks, suggesting that resting-state calibration reduces the amount of supervised task data required for subject-specific adaptation.

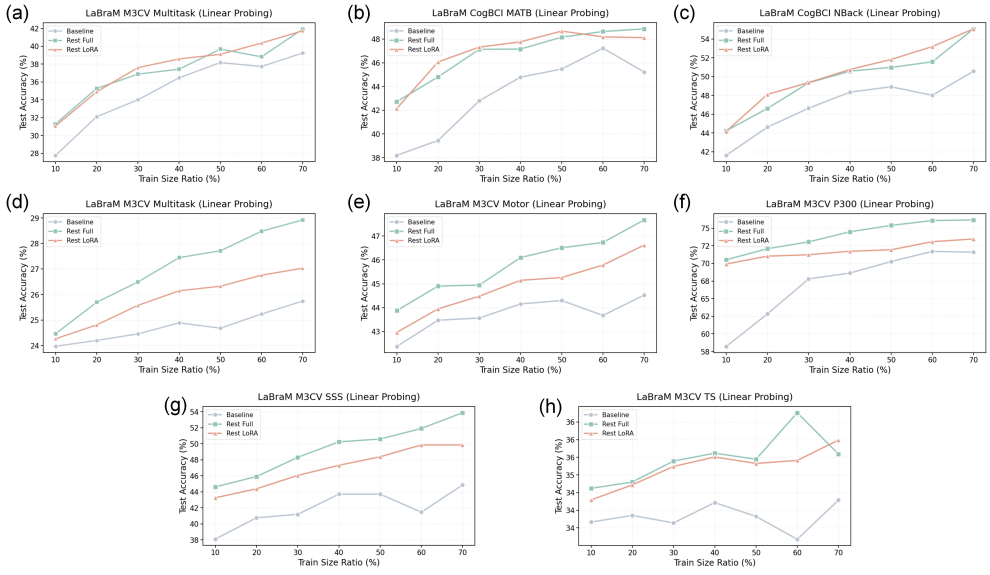


Figure A2: Data efficiency curves across downstream tasks of LaBraM model

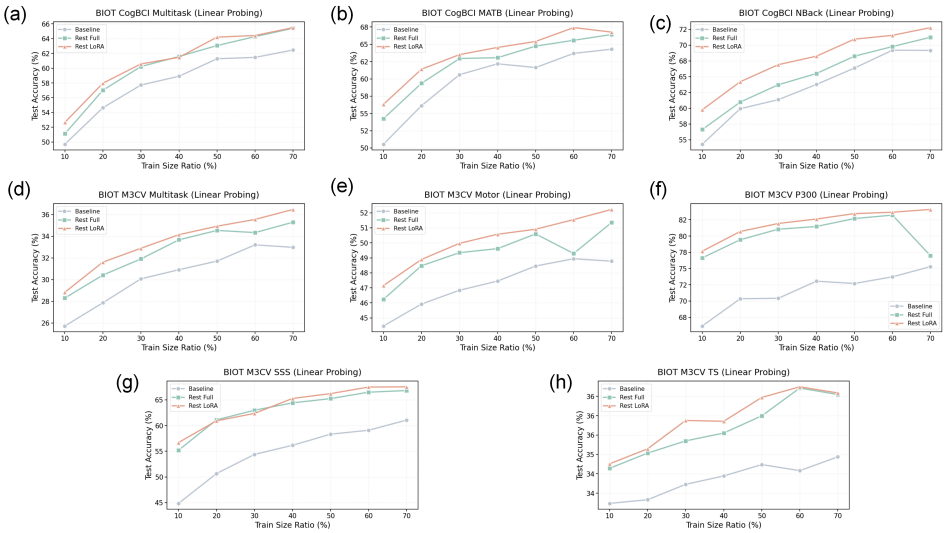


Figure A3: Data efficiency curves across downstream tasks of BIOT model

I LoRA Adapter Process Diagram

In this section, we present an analysis of task-to-task interpolation, where we examine the interpolation curves between Task1→Task2 and Task2→Task1. The average performance across both directions is reflected in the combined accuracy curve shown in Figure A4. The results highlight the performance landscape of LoRA adapters in the context of multi-task adaptation.

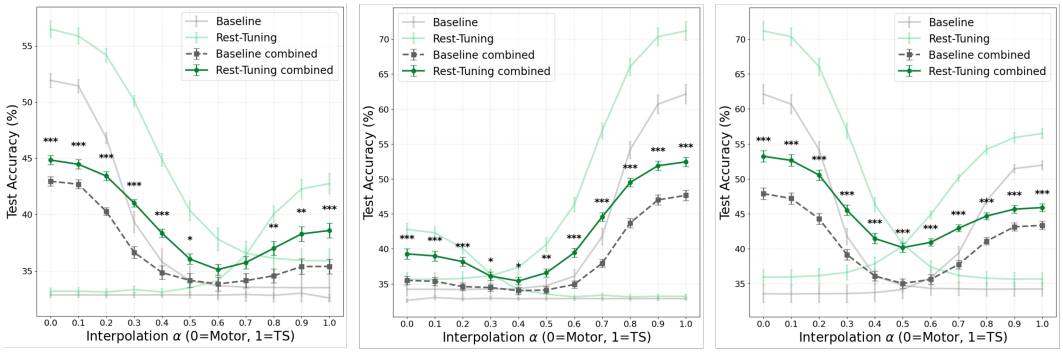


Figure A4: Detailed LoRA adapter interpolation between two tasks. Rest-Tuning (green) maintains more stable accuracy along the interpolation path compared to task-only LoRA (gray dashed).

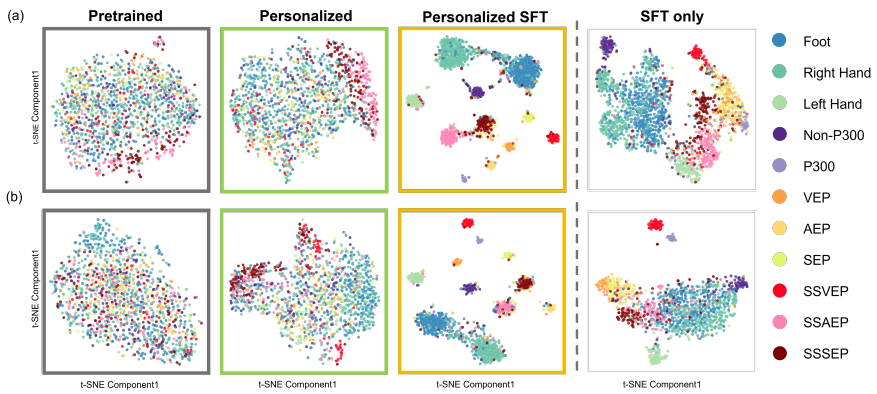


Figure A5: t-SNE visualization of LaBraM embeddings under multitask settings. We analyze two subjects using Rest-Full and SFT with 70% labeled data.

J T-SNE visualization of embeddings

We present additional t-SNE visualizations of task embeddings. Figure A5 shows multitask embedding results obtained with the LaBraM backbone on the M3CV dataset, which is analyzed for a single subject under the Rest-Full and SFT settings with 70% labeled data.

K Rest Control

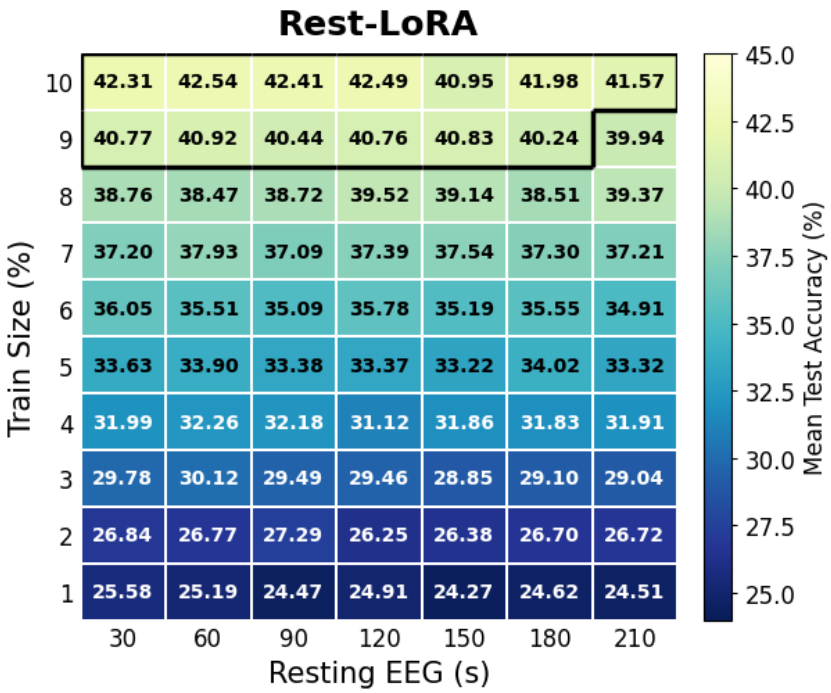


Figure A6: **Resting EEG duration vs. multitask accuracy.** LabraM Rest-LoRA performance rapidly saturates with 1–3 minutes of resting EEG.

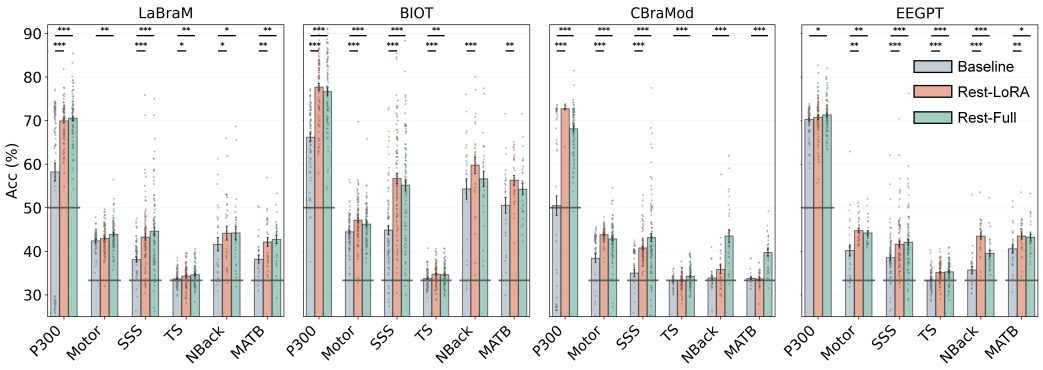


Figure A7: **Linear probing performance under 10% training data.** Bars show mean accuracy across subjects for four EEG FMs and six downstream tasks. Gray denotes the no-rest baseline, orange denotes Rest-LoRA, and green denotes Rest-Full. Error bars indicate standard deviation; significance is computed with paired subject-level tests comparing Rest-Tuning to the no-rest baseline.

L Detail Performance Results

Performance under 10% and 70% training samples. We first evaluate Rest-Tuning under a low-label setting where only 10% of the task training samples are used. As shown in Figure A7, both Rest-LoRA and Rest-Full improve linear probing performance over the no-rest baseline across most EEG foundation models and downstream tasks. Table A9 further shows that these improvements persist under both uni-task and multi-task adaptation on M3CV and COGBCI. These results suggest that Rest-Tuning is especially beneficial when labeled task data are limited, providing a subject-calibrated backbone that improves downstream adaptation with minimal supervision.

We also report the corresponding results under the 70% training-sample setting in Table 1. This setting provides a higher-label regime for evaluating whether the benefits of Rest-Tuning remain when more supervised task data are available. Rest-Tuning continues to improve downstream performance across different EEG foundation models, datasets, and adaptation protocols, indicating that resting-state calibration remains useful beyond the extremely low-label setting.

Detailed uni-task performance profiles. Figures A8 and A9 provide detailed uni-task performance profiles for each EEG foundation model on M3CV under 10% and 70% training data, respectively. The radar plots compare task-only adaptation with Rest-Tuning followed by task-specific SFT across six downstream tasks. Overall, Rest-Tuning improves the normalized task-level profiles for most backbones and adaptation settings, showing that the subject-calibrated backbone provides consistent benefits beyond the aggregate results reported in the main tables.

Table A9: **Uni-task and multi-task** performance on M3CV and COG-BCI datasets under 10% training data.

EEG Foundation Model	Rest-Tuning	Task-SFT	Trainable Params (Rest/Task)	M3CV				COGBCI			
				Unitask		Multitask		Unitask		Multitask	
				$\overline{\text{ACC}}$(%)	ΔACC(%)	$\overline{\text{ACC}}$(%)	ΔACC(%)	$\overline{\text{ACC}}$(%)	ΔACC(%)	$\overline{\text{ACC}}$(%)	ΔACC(%)
LaBraM	/	Task-LoRA	0 / 5.00%	49.36	/	25.39	/	48.66	/	37.50	/
	Rest-LoRA	Task-LoRA	5% / 5%	52.41	↑3.05	26.35	↑0.96	49.22	↑0.56	39.78	↑2.28
	Rest-Full	Task-LoRA	100% / 5%	53.13	↑3.77	26.93	↑1.54	50.06	↑1.40	40.42	↑2.92
	/	Task-Full	0 / 100%	50.81	/	26.12	/	49.48	/	39.41	/
	Rest-Full	Task-Full	100% / 100%	54.80	↑3.99	28.08	↑1.96	51.05	↑1.57	44.42	↑5.01
BIOT	/	Task-LoRA	0 / 4.5%	55.13	/	33.34	/	77.61	/	75.80	/
	Rest-LoRA	Task-LoRA	4.5% / 4.5%	61.04	↑5.91	34.19	↑0.85	79.97	↑2.36	77.69	↑1.89
	Rest-Full	Task-LoRA	100% / 4.5%	60.69	↑5.56	33.40	↑0.06	79.31	↑1.70	77.24	↑1.44
	/	Task-Full	0 / 100%	58.22	/	35.62	/	80.02	/	77.45	/
	Rest-Full	Task-Full	100% / 100%	62.29	↑4.07	36.35	↑0.73	81.37	↑1.35	78.59	↑1.14
CBraMod	/	Task-LoRA	0 / 3.03%	44.94	/	23.95	/	38.16	/	22.74	/
	Rest-LoRA	Task-LoRA	3.03% / 3.03%	47.88	↑2.94	24.50	↑0.55	46.22	↑8.06	26.00	↑3.26
	Rest-Full	Task-LoRA	100% / 3.03%	49.45	↑4.51	24.64	↑0.69	52.58	↑14.42	26.00	↑3.26
	/	Task-Full	0 / 100%	51.07	/	25.66	/	55.30	/	46.06	/
	Rest-Full	Task-Full	100% / 100%	53.12	↑2.05	26.08	↑0.42	57.96	↑2.66	50.18	↑4.12
EEGPT	/	Task-LoRA	0 / 3.03%	48.48	/	16.90	/	41.80	/	32.26	/
	Rest-LoRA	Task-LoRA	3.98% / 3.98%	49.26	↑0.78	17.78	↑0.88	42.44	↑0.64	32.54	↑0.28
	Rest-Full	Task-LoRA	100% / 3.98%	49.18	↑0.70	18.74	↑1.84	43.56	↑1.76	33.76	↑1.50
	/	Task-Full	0 / 100%	48.98	/	17.88	/	42.44	/	32.91	/
	Rest-Full	Task-Full	100% / 100%	49.56	↑0.58	19.14	↑1.26	43.33	↑0.89	34.35	↑1.44

Table A10: **Uni-task and multi-task** performance on M3CV and COG-BCI datasets.

EEG foundation model	Rest-Tuning	Task-SFT	Unitask								Multitask							
			M3CV				COG-BCI				M3CV				COG-BCI			
			train 10%		train 70%		train 10%		train 70%		train 10%		train 70%		train 10%		train 70%	
			$\overline{\text{F1}}$ (%)	Δ F1(%)	$\overline{\text{F1}}$ (%)	Δ F1(%)	$\overline{\text{F1}}$ (%)	Δ F1(%)	$\overline{\text{F1}}$ (%)	Δ F1(%)	F1 (%)	Δ F1(%)	F1 (%)	Δ F1(%)	F1 (%)	Δ F1(%)	F1 (%)	Δ F1(%)
LaBraM	/	LoRA	44.92	/	55.24	/	47.92	/	59.86	/	19.68	/	34.17	/	36.84	/	56.39	/
	Rest-LoRA	LoRA	51.10	↑6.18	63.99	↑8.75	48.39	↑0.47	63.14	↑3.28	21.46	↑1.78	40.33	↑6.16	39.18	↑2.34	60.92	↑4.53
	Rest-Full	LoRA	52.00	↑7.08	64.35	↑9.11	49.48	↑1.56	68.70	↑8.84	22.22	↑2.54	41.23	↑7.06	39.68	↑2.84	66.53	↑10.14
	/	Full	48.47	/	59.97	/	48.40	/	72.19	/	21.29	/	44.23	/	38.68	/	73.24	/
	Rest-Full	Full	53.59	↑5.12	67.90	↑7.93	50.42	↑2.02	75.30	↑3.11	23.69	↑2.40	48.44	↑4.21	43.62	↑4.94	77.95	↑4.71
BIOT	/	LoRA	53.04	/	64.87	/	77.35	/	90.08	/	30.45	/	48.08	/	75.60	/	88.93	/
	Rest-LoRA	LoRA	60.62	↑7.58	70.35	↑5.48	79.86	↑2.51	90.92	↑0.84	31.73	↑1.28	50.87	↑2.79	77.53	↑1.93	89.46	↑0.53
	Rest-Full	LoRA	59.61	↑6.57	67.63	↑2.76	79.17	↑1.82	91.19	↑1.11	31.25	↑0.80	49.21	↑1.13	77.06	↑1.46	89.58	↑0.65
	/	Full	56.23	/	66.82	/	79.92	/	90.38	/	32.97	/	49.32	/	77.24	/	89.33	/
	Rest-Full	Full	61.93	↑5.70	66.85	↑0.03	81.25	↑1.33	91.23	↑0.85	34.40	↑1.43	51.45	↑2.13	78.38	↑1.14	90.44	↑1.11
CBraMod	/	LoRA	37.25	/	40.98	/	34.81	/	45.76	/	12.40	/	14.51	/	19.36	/	36.98	/
	Rest-LoRA	LoRA	41.36	↑4.11	46.96	↑5.98	44.39	↑9.58	59.62	↑13.86	14.36	↑1.96	16.17	↑1.66	25.54	↑6.18	39.71	↑2.73
	Rest-Full	LoRA	45.38	↑8.13	54.04	↑13.06	51.81	↑17.00	64.94	↑19.18	15.58	↑3.18	20.42	↑5.91	25.54	↑6.18	47.93	↑10.95
	/	Full	43.99	/	56.93	/	52.89	/	70.71	/	15.36	/	30.60	/	42.21	/	70.23	/
	Rest-Full	Full	52.02	↑8.03	62.05	↑5.12	57.36	↑4.47	75.24	↑4.53	20.20	↑4.84	35.67	↑5.07	49.16	↑6.95	73.44	↑3.21
EEGPT	/	LoRA	44.58	/	49.38	/	41.39	/	54.39	/	13.89	/	25.99	/	32.14	/	48.28	/
	Rest-LoRA	LoRA	47.53	↑2.95	52.79	↑3.41	42.30	↑0.91	54.70	↑0.31	15.16	↑1.27	26.19	↑0.20	32.28	↑0.14	48.58	↑0.30
	Rest-Full	LoRA	47.49	↑2.91	54.57	↑5.19	43.49	↑2.10	54.63	↑0.24	16.09	↑2.20	26.88	↑0.89	33.41	↑1.27	49.67	↑1.39
	/	Full	45.42	/	52.43	/	41.78	/	54.25	/	15.65	/	25.30	/	32.80	/	47.60	/
	Rest-Full	Full	48.08	↑2.66	57.88	↑5.45	42.90	↑1.12	57.06	↑2.81	16.92	↑1.27	28.69	↑3.39	34.01	↑1.21	57.67	↑10.07

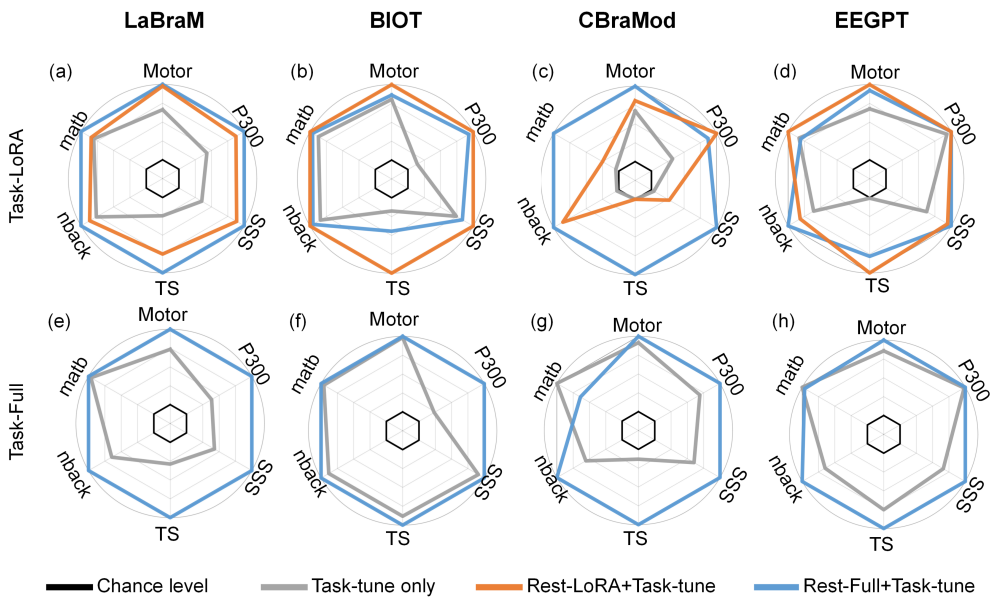


Figure A8: **Uni-task performance profiles on M3CV under 10% training data.** We display normalized accuracy across six tasks using Rest-Tuning on four different EEG FM backbones. Task-specific SFT adaptation strategies, including LoRA and Full tuning, are used after Rest-Tuning.

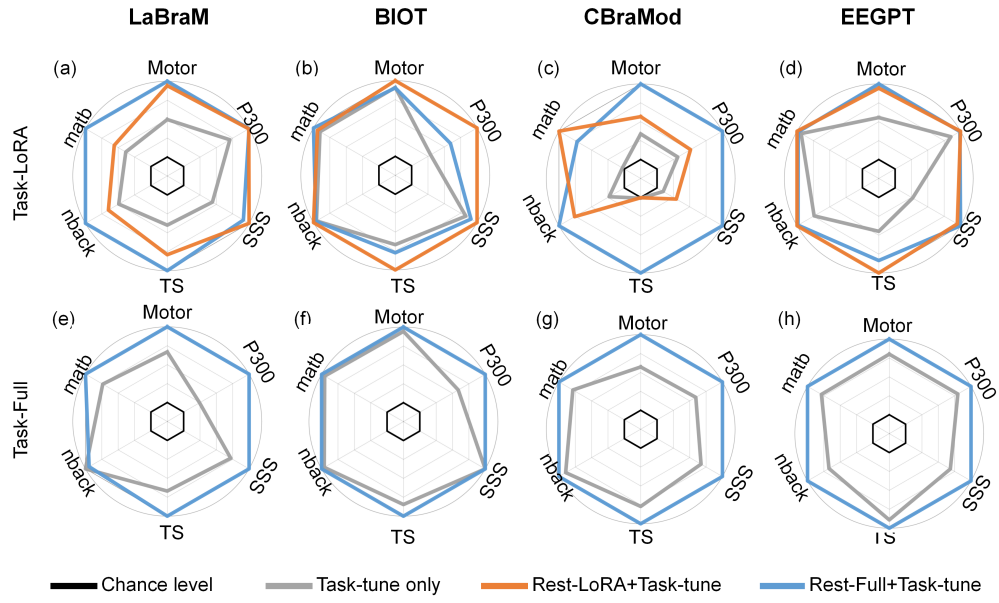


Figure A9: **Uni-task performance profiles on M3CV under 70% training data.** We display normalized accuracy across six tasks using Rest-Tuning on four different EEG FM backbones.

Table A11: Uni-task performance on M3CV and COG-BCI datasets (training size = 0.1)

EEG foundation model	Rest-Tuning	Task-SFT	**Motor** (3-Class)		**P300** (2-Class)		**Steady State Sensory** (3-Class)		**Transient Sensory** (3-Class)		**Nback** (3-Class)		**MATB** (3-Class)	
			ACC	**F1**	**ACC**	**F1**	**ACC**	**F1**	**ACC**	**F1**	**ACC**	**F1**	**ACC**	**F1**
LaBraM	/	Task-LP	0.4238±0.0380	0.3348±0.0657	0.5821±0.1936	0.4884±0.2273	0.3811±0.0570	0.2773±0.0786	0.3367±0.0172	0.2724±0.0629	0.4161±0.0824	0.3867±0.0985	0.3819±0.0493	0.3326±0.0962
	Rest-LoRA	Task-LP	0.4296±0.0297	0.3994±0.0341	0.6991±0.0469	0.6349±0.0454	0.4317±0.0763	0.3883±0.0835	0.3429±0.0239	0.3226±0.0329	0.4415±0.0869	0.4335±0.0884	0.4214±0.0554	0.4109±0.0600
	Rest-Full	Task-LP	0.4388±0.0339	0.4070±0.0384	0.6497±0.1121	0.6123±0.1099	0.4461±0.0883	0.4043±0.0940	0.3462±0.0234	0.3291±0.0341	0.4422±0.0813	0.4263±0.0835	0.4272±0.0506	0.4220±0.0530
	/	Task-LoRA	0.4473±0.0268	0.4076±0.0545	0.7099±0.0391	0.6302±0.0361	0.4696±0.0938	0.4404±0.1080	0.3474±0.0236	0.3184±0.0520	0.5089±0.1088	0.5038±0.1074	0.4642±0.0542	0.4546±0.0622
	Rest-LoRA	Task-LoRA	0.4629±0.0368	0.4485±0.0437	0.7663±0.0656	0.7510±0.0759	0.5132±0.1082	0.5007±0.1132	0.3538±0.0242	0.3438±0.0284	0.5163±0.1049	0.5099±0.1020	0.4681±0.0421	0.4578±0.0423
	Rest-Full	Task-LoRA	0.4640±0.0386	0.4508±0.0425	0.7785±0.0644	0.7651±0.0741	0.5268±0.1032	0.5153±0.1075	0.3559±0.0262	0.3486±0.0269	0.5253±0.1006	0.5183±0.0999	0.4759±0.0616	0.4712±0.0619
	/	Task-Full	0.4618±0.0371	0.4235±0.0617	0.7251±0.0293	0.7333±0.0870	0.4859±0.0988	0.4539±0.1196	0.3596±0.0271	0.3278±0.0490	0.5212±0.1097	0.5134±0.1135	0.4684±0.0536	0.4546±0.0593
	Rest-Full	Task-Full	0.4758±0.0412	0.4620±0.0458	0.8096±0.0796	0.7959±0.0901	0.5358±0.1192	0.5232±0.1239	0.3708±0.0289	0.3626±0.0311	0.5509±0.1157	0.5461±0.1165	0.4701±0.0654	0.4623±0.0657
BIOT	/	Task-LP	0.4446±0.0375	0.4154±0.0541	0.6616±0.0818	0.6240±0.0684	0.4485±0.0965	0.4084±0.1168	0.3373±0.0199	0.3081±0.0442	0.5430±0.1249	0.5243±0.1309	0.5055±0.0967	0.4810±0.1211
	Rest-LoRA	Task-LP	0.4715±0.0507	0.4619±0.0532	0.7767±0.0820	0.7612±0.0901	0.5669±0.1140	0.5573±0.1183	0.3476±0.0230	0.3412±0.0248	0.5976±0.1045	0.5896±0.1044	0.5631±0.0582	0.5553±0.0592
	Rest-Full	Task-LP	0.4624±0.0437	0.4543±0.0471	0.7663±0.0808	0.7568±0.0852	0.5518±0.1179	0.5438±0.1219	0.3464±0.0235	0.3403±0.0257	0.5663±0.0941	0.5550±0.0990	0.5425±0.0697	0.5354±0.0707
	/	Task-LoRA	0.5063±0.0478	0.4975±0.0556	0.6965±0.0631	0.6518±0.0697	0.6561±0.1246	0.6497±0.1303	0.3464±0.0259	0.3225±0.0508	0.7947±0.1031	0.7912±0.1055	0.7574±0.0606	0.7558±0.0615
	Rest-LoRA	Task-LoRA	0.5192±0.0459	0.5153±0.0484	0.8757±0.0653	0.8726±0.0691	0.6900±0.1083	0.6868±0.1122	0.3566±0.0242	0.3501±0.0250	0.8220±0.0936	0.8217±0.0935	0.7773±0.0686	0.7754±0.0691
	Rest-Full	Task-LoRA	0.5103±0.0497	0.5063±0.0525	0.8672±0.0746	0.8650±0.0755	0.6690±0.1111	0.6656±0.1137	0.3511±0.0238	0.3474±0.0242	0.8146±0.0944	0.8138±0.0950	0.7715±0.0745	0.7696±0.0743
	/	Task-Full	0.5308±0.0488	0.5229±0.0564	0.7382±0.0706	0.7002±0.0872	0.7031±0.1247	0.6951±0.1356	0.3567±0.0242	0.3311±0.0450	0.8104±0.0876	0.8107±0.0881	0.7899±0.0569	0.7877±0.0582
	Rest-Full	Task-Full	0.5319±0.0465	0.5299±0.0479	0.8871±0.0635	0.8835±0.0681	0.7150±0.1130	0.7124±0.1147	0.3577±0.0224	0.3513±0.0258	0.8309±0.0936	0.8301±0.0940	0.7965±0.0664	0.7949±0.0666
CBraMod	/	Task-LP	0.3839±0.0980	0.2543±0.0820	0.5048±0.2180	0.3857±0.24105	0.3501±0.0796	0.2155±0.0785	0.3325±0.0098	0.1990±0.0439	0.3389±0.0271	0.2085±0.0520	0.3379±0.0154	0.2092±0.0522
	Rest-LoRA	Task-LP	0.4381±0.0157	0.3856±0.0362	0.7269±0.0094	0.6141±0.0063	0.4076±0.0348	0.3384±0.0440	0.3328±0.0170	0.2897±0.0298	0.3586±0.0551	0.3011±0.0656	0.3392±0.0221	0.2991±0.0423
	Rest-Full	Task-LP	0.4283±0.0407	0.3968±0.0419	0.6811±0.0550	0.6257±0.0450	0.4314±0.0893	0.3885±0.0954	0.3428±0.0240	0.3146±0.0349	0.4675±0.0947	0.4374±0.1018	0.4009±0.0469	0.3749±0.0532
	/	Task-LoRA	0.4334±0.0134	0.3406±0.0523	0.6556±0.0970	0.5960±0.0719	0.3749±0.0453	0.2935±0.0606	0.3335±0.0128	0.2600±0.0468	0.3994±0.0730	0.3651±0.0915	0.3637±0.0333	0.3310±0.0430
	Rest-LoRA	Task-LoRA	0.4397±0.0291	0.3902±0.0456	0.7264±0.0162	0.6140±0.0091	0.4140±0.0485	0.3500±0.0554	0.3351±0.0151	0.3000±0.0341	0.5377±0.1565	0.5220±0.1667	0.3866±0.0685	0.3657±0.0675
	Rest-Full	Task-LoRA	0.4475±0.0406	0.4135±0.0472	0.7159±0.0652	0.6476±0.0751	0.4575±0.1189	0.4140±0.1222	0.3570±0.0394	0.3401±0.0477	0.5730±0.1430	0.5653±0.1494	0.4785±0.0937	0.4708±0.0957
	/	Task-Full	0.4597±0.0370	0.3962±0.0631	0.7366±0.0376	0.6365±0.0643	0.4956±0.0974	0.4332±0.1188	0.3510±0.0244	0.2937±0.0496	0.5654±0.1206	0.5454±0.1253	0.5406±0.0924	0.5123±0.1029
	Rest-Full	Task-Full	0.4635±0.0415	0.4500±0.0466	0.7677±0.0722	0.7566±0.0797	0.5256±0.1021	0.5149±0.1071	0.3680±0.0236	0.3593±0.0254	0.6329±0.1195	0.6279±0.1222	0.5262±0.0902	0.5192±0.0910
EEGPT	/	Task-LP	0.4019±0.0671	0.3899±0.0813	0.7028±0.0304	0.6187±0.0170	0.3854±0.0597	0.3335±0.0734	0.3338±0.0216	0.2818±0.0561	0.3572±0.0445	0.3392±0.0623	0.4058±0.0484	0.3951±0.0670
	Rest-LoRA	Task-LP	0.4445±0.0291	0.4168±0.0338	0.7075±0.0501	0.6467±0.0494	0.4165±0.0623	0.3865±0.0652	0.3514±0.0216	0.3482±0.0223	0.3820±0.0376	0.3782±0.0403	0.4349±0.0389	0.4333±0.0391
	Rest-Full	Task-LP	0.4408±0.0264	0.4156±0.0300	0.7133±0.0369	0.6563±0.0462	0.4210±0.0673	0.3954±0.0683	0.3528±0.0229	0.3496±0.0250	0.3957±0.0392	0.3913±0.0416	0.4325±0.0379	0.4296±0.0355
	/	Task-LoRA	0.4560±0.0693	0.4530±0.0733	0.7219±0.0376	0.6466±0.0548	0.4226±0.0732	0.3803±0.0860	0.3383±0.0259	0.3032±0.0452	0.3868±0.0409	0.3791±0.0445	0.4492±0.0532	0.4486±0.0537
	Rest-LoRA	Task-LoRA	0.4492±0.0315	0.4306±0.0359	0.7270±0.0445	0.6994±0.0555	0.4359±0.0699	0.4173±0.0760	0.3582±0.0247	0.3538±0.0261	0.3922±0.0321	0.3903±0.0330	0.4565±0.0376	0.4557±0.0372
	Rest-Full	Task-LoRA	0.4461±0.0333	0.4299±0.0377	0.7270±0.0445	0.6974±0.0556	0.4378±0.0755	0.4201±0.0813	0.3562±0.0245	0.3521±0.0257	0.4080±0.0382	0.4065±0.0390	0.4632±0.0341	0.4633±0.0337
	/	Task-Full	0.4449±0.0272	0.4017±0.0515	0.7297±0.0501	0.6771±0.0699	0.4280±0.0658	0.3980±0.0775	0.3564±0.0274	0.3400±0.0392	0.3979±0.0370	0.3873±0.0387	0.4508±0.0413	0.4483±0.0418
	Rest-Full	Task-Full	0.4508±0.0328	0.4369±0.0370	0.7311±0.0504	0.7056±0.0571	0.4419±0.0696	0.4299±0.0731	0.3587±0.0244	0.3508±0.0280	0.4059±0.0383	0.4029±0.0389	0.4607±0.0369	0.4550±0.0399

Table A12: Uni-task performance on M3CV and COG-BCI datasets (training size = 70%)

EEG foundation model	Rest-Tuning	Task-SFT	**Motor** (3-Class)		**P300** (2-Class)		**Steady State Sensory** (3-Class)		**Transient Sensory** (3-Class)		**Nback** (3-Class)		**MATB** (3-Class)	
			ACC	**F1**	**ACC**	**F1**	**ACC**	**F1**	**ACC**	**F1**	**ACC**	**F1**	**ACC**	**F1**
LaBraM	/	Task-LP	0.4453±0.0456	0.3913±0.0651	0.7160±0.0219	0.6211±0.0274	0.4485±0.1004	0.3840±0.1124	0.3429±0.0390	0.3165±0.0621	0.5056±0.1143	0.4976±0.1151	0.4522±0.0534	0.4891±0.0556
	Rest-LoRA	Task-LP	0.4662±0.0443	0.4389±0.0514	0.7347±0.0621	0.6647±0.0824	0.4908±0.1018	0.4588±0.1075	0.3599±0.0418	0.3487±0.0440	0.5508±0.1058	0.5456±0.1054	0.4812±0.0542	0.4715±0.0528
	Rest-Full	Task-LP	0.4767±0.0546	0.4556±0.0636	0.7618±0.0748	0.7123±0.0976	0.5330±0.1146	0.5073±0.1234	0.3559±0.0434	0.3439±0.0471	0.5507±0.0960	0.5410±0.0995	0.4886±0.0561	0.4803±0.0574
	/	Task-LoRA	0.5198±0.0616	0.5111±0.0664	0.7593±0.0597	0.7151±0.0868	0.6210±0.1359	0.6156±0.1385	0.3783±0.0458	0.3676±0.0506	0.6479±0.1091	0.6428±0.1108	0.5613±0.0745	0.5543±0.0640
	Rest-LoRA	Task-LoRA	0.5644±0.0728	0.5617±0.0737	0.9078±0.0620	0.9059±0.0646	0.7117±0.1298	0.7078±0.1321	0.3898±0.0438	0.3843±0.0459	0.6786±0.1127	0.6746±0.1142	0.5919±0.0741	0.5881±0.0777
	Rest-Full	Task-LoRA	0.5690±0.0714	0.5654±0.0745	0.9084±0.0583	0.9060±0.0618	0.7176±0.1264	0.7148±0.1286	0.3944±0.0486	0.3878±0.0499	0.7319±0.1166	0.7294±0.1172	0.6469±0.0863	0.6445±0.0879
	/	Task-Full	0.5765±0.0691	0.5662±0.0785	0.7792±0.0713	0.7457±0.0988	0.7100±0.1412	0.7040±0.1464	0.4008±0.0433	0.3828±0.0550	0.7636±0.1232	0.7574±0.1277	0.6879±0.0835	0.6864±0.0834
	Rest-Full	Task-Full	0.6114±0.0741	0.6074±0.0768	0.9472±0.0508	0.9457±0.0545	0.7537±0.1301	0.7482±0.1356	0.4222±0.0461	0.4146±0.0492	0.7844±0.1281	0.7823±0.1291	0.7260±0.1123	0.7236±0.1141
BIOT	/	Task-LP	0.4829±0.0546	0.4680±0.0611	0.7528±0.1185	0.7318±0.1296	0.5683±0.1310	0.5530±0.1460	0.3428±0.0402	0.3241±0.0568	0.6583±0.1080	0.6503±0.1111	0.6234±0.0686	0.6174±0.0711
	Rest-LoRA	Task-LP	0.5223±0.0562	0.5166±0.0590	0.8409±0.0675	0.8288±0.0804	0.6754±0.1084	0.6704±0.1109	0.3659±0.0410	0.3594±0.0423	0.7277±0.0926	0.7239±0.0939	0.6676±0.0621	0.6623±0.0661
	Rest-Full	Task-LP	0.5136±0.0621	0.5096±0.0642	0.7698±0.1319	0.7578±0.1335	0.6800±0.1116	0.6630±0.1141	0.3655±0.0411	0.3611±0.0418	0.7124±0.1046	0.7078±0.1058	0.6641±0.0671	0.6587±0.0709
	/	Task-LoRA	0.6022±0.0630	0.5973±0.0657	0.7947±0.0867	0.7812±0.0976	0.8463±0.0870	0.8454±0.0879	0.3793±0.0406	0.3710±0.0421	0.9148±0.0588	0.9145±0.0591	0.8880±0.0459	0.8871±0.0466
	Rest-LoRA	Task-LoRA	0.6121±0.0631	0.6109±0.0640	0.9451±0.0427	0.9441±0.0440	0.8795±0.0806	0.8786±0.0816	0.3860±0.0491	0.3803±0.0498	0.9238±0.0457	0.9234±0.0462	0.8958±0.0526	0.8949±0.0541
	Rest-Full	Task-LoRA	0.6029±0.0662	0.6015±0.0676	0.8740±0.0940	0.8667±0.1026	0.8620±0.0885	0.8613±0.0892	0.3817±0.0392	0.3755±0.0401	0.9165±0.0574	0.9163±0.0578	0.9077±0.0453	0.9075±0.0455
	/	Task-Full	0.6158±0.0679	0.6132±0.0686	0.8306±0.0802	0.8239±0.0867	0.8712±0.0841	0.8698±0.0859	0.3807±0.0423	0.3660±0.0491	0.9128±0.0567	0.9130±0.0565	0.8950±0.0526	0.8945±0.0526
	Rest-Full	Task-Full	0.6214±0.0629	0.6198±0.0638	0.8947±0.0879	0.8874±0.0979	0.8713±0.0884	0.8706±0.0892	0.3835±0.0355	0.3763±0.0375	0.9208±0.0516	0.9206±0.0519	0.9040±0.0421	0.9039±0.0419
CBraMod	/	Task-LP	0.4200±0.0537	0.2838±0.0550	0.6097±0.1756	0.5153±0.1852	0.3881±0.0610	0.2567±0.0744	0.3367±0.0180	0.2097±0.0501	0.3742±0.0376	0.3028±0.0653	0.3386±0.0213	0.2216±0.0537
	Rest-LoRA	Task-LP	0.4384±0.0265	0.3836±0.0434	0.7279±0.0021	0.6132±0.0028	0.4138±0.0614	0.3491±0.0629	0.3330±0.0275	0.2642±0.0416	0.3827±0.0606	0.3194±0.0619	0.3405±0.0256	0.2840±0.0370
	Rest-Full	Task-LP	0.4557±0.0472	0.4258±0.0524	0.7070±0.0558	0.6414±0.0614	0.4816±0.0845	0.4425±0.0928	0.3589±0.0390	0.3444±0.0418	0.5267±0.0877	0.5069±0.0897	0.4479±0.0669	0.4281±0.0675
	/	Task-LoRA	0.4376±0.0227	0.3586±0.0528	0.7089±0.0375	0.6134±0.0181	0.4298±0.0581	0.3539±0.0699	0.3444±0.0357	0.3132±0.0496	0.5511±0.1511	0.5369±0.1626	0.4041±0.0936	0.3783±0.1003
	Rest-LoRA	Task-LoRA	0.4571±0.0699	0.4297±0.0716	0.7451±0.0969	0.6954±0.1190	0.4745±0.1061	0.4355±0.1206	0.3417±0.0377	0.3176±0.0443	0.6567±0.1295	0.6452±0.1316	0.5622±0.1156	0.5472±0.1184
	Rest-Full	Task-LoRA	0.4829±0.0661	0.4690±0.0690	0.8064±0.0781	0.7964±0.0863	0.5454±0.1206	0.5237±0.1284	0.3833±0.0561	0.3726±0.0561	0.6754±0.1252	0.6675±0.1258	0.5447±0.0998	0.5313±0.1012
	/	Task-Full	0.4957±0.0569	0.4641±0.0749	0.8335±0.0655	0.8257±0.0756	0.6295±0.1175	0.6114±0.1320	0.3967±0.0474	0.3758±0.0620	0.7686±0.0994	0.7630±0.1026	0.6674±0.0834	0.6512±0.0960
	Rest-Full	Task-Full	0.5288±0.0662	0.5236±0.0673	0.8970±0.0659	0.8966±0.0655	0.6713±0.1241	0.6660±0.1286	0.4024±0.0489	0.3957±0.0513	0.8046±0.0933	0.8035±0.0938	0.7033±0.0796	0.7012±0.0802
EEGPT	/	Task-LP	0.4417±0.0428	0.4004±0.0527	0.7226±0.0361	0.6444±0.0537	0.4168±0.0763	0.3676±0.0793	0.3498±0.0410	0.3323±0.0534	0.4793±0.0669	0.4760±0.0674	0.5054±0.0487	0.5017±0.0484
	Rest-LoRA	Task-LP	0.4712±0.0561	0.4545±0.0624	0.7656±0.0590	0.7280±0.0804	0.4904±0.0942	0.4740±0.1023	0.3774±0.0443	0.3731±0.0454	0.5412±0.0565	0.5362±0.0597	0.5432±0.0604	0.5413±0.0607
	Rest-Full	Task-LP	0.4670±0.0561	0.4503±0.0628	0.7727±0.0679	0.7411±0.0829	0.4978±0.0988	0.4852±0.1037	0.3778±0.0415	0.3744±0.0431	0.5282±0.0547	0.5233±0.0560	0.5235±0.0599	0.5179±0.0624
	/	Task-LoRA	0.4596±0.0572	0.4283±0.0678	0.7895±0.0650	0.7681±0.0842	0.4613±0.1100	0.4220±0.1236	0.3707±0.0444	0.3566±0.0578	0.5392±0.0834	0.5360±0.0828	0.5562±0.0590	0.5518±0.0619
	Rest-LoRA	Task-LoRA	0.5544±0.1094	0.5402±0.1158	0.7882±0.0791	0.7710±0.0896	0.4328±0.1175	0.4051±0.1211	0.4063±0.0818	0.3953±0.0875	0.5541±0.0887	0.5528±0.0887	0.5443±0.0607	0.5411±0.0616
	Rest-Full	Task-LoRA	0.5612±0.1111	0.5499±0.1157	0.8070±0.0717	0.7914±0.0820	0.4664±0.1237	0.4326±0.1306	0.4173±0.0839	0.4088±0.0849	0.5423±0.0822	0.5395±0.0841	0.5574±0.0579	0.5531±0.0590
	/	Task-Full	0.4783±0.0495	0.4536±0.0612	0.7942±0.0663	0.7804±0.0766	0.5125±0.0980	0.4972±0.1057	0.3804±0.0485	0.3658±0.0571	0.5442±0.0849	0.5373±0.0859	0.5568±0.0653	0.5477±0.0736
	Rest-Full	Task-Full	0.4894±0.0577	0.4812±0.0613	0.8171±0.0686	0.5672±0.1097	0.5513±0.1136	0.8004±0.0669	0.7862±0.0782	0.4663±0.1203	0.4387±0.1262	0.4230±0.0787	0.4122±0.0837	0.5894±0.1084

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